PRODUCT REQUIREMENTS DOCUMENT

Luggik.

Logistics-Anchored Escrow Infrastructure: Converting Physical Delivery Carriers into Smart Real-Time Financial Oracles

NOMBA FOCUS TRACK	TECHNICAL STACK	HACKATHON WINDOW
Virtual Accounts as Infrastructure	**Node.js, PostgreSQL, Nomba APIs**	**Sprint: July 1 - July 7, 2026**

Executive Summary

In the rapidly expanding Nigerian digital economy, social commerce (Instagram, WhatsApp, and Twitter marketplaces) represents billions of Naira in monthly transactional volume. However, this ecosystem is throttled by a fundamental, systemic friction: **the trust deadlock**. Buyers are terrified of getting scammed ("pay-before-delivery"), while vendors face unsustainable financial losses or security risks from "payment-on-delivery" dropouts.

Traditional escrow solutions fail due to extreme operational overhead—forcing both parties through complex application signups, manual contract generation, and messy customer support dispute review loops relying on unverified chat screenshots.

> **THE LUGGIK STRUCTURAL INNOVATION**
>
> Luggik completely eliminates the dispute layer by converting verified logistics companies into automated, physical API oracles. Funds are securely locked upfront via **Nomba Dedicated Virtual Accounts**, but are only liquidated and split via the **Nomba Transfers API** when the delivery network's courier validates the inventory physically. No human arbiters, no chat logs, no adoption friction.

Product Use Case & Core Dynamics

Luggik seamlessly sits inside the traditional merchant-buyer interaction without requiring either party to change their daily behavior, register accounts, or install a separate mobile application.

The Operational Flow:

1. **Frictionless Purchase Initiation**

   A consumer spots an item on Instagram. On Luggik's ultra-lightweight web checkout page, they input the vendor's phone number, the item price, and select their preferred logistics provider (e.g., GIGL, Topship).

2. **Dynamic Account Provisioning**

   Luggik calls Nomba's API to dynamically spawn an isolated Temporary Virtual Account branded explicitly for that transaction (e.g., *"Luggik / Lagos Bags"*). The buyer transfers the total sum (Item + Delivery Fee) directly via their standard bank app.

3. **Headless Vendor Notification**

   The moment the funds land, Nomba's transaction webhook hits Luggik. Luggik automatically dispatches a verified WhatsApp notification to the vendor showing the live, institutional vaulting of their funds, prompting immediate fulfillment.

4 **The Physical Oracle Point**

The logistics driver arrives at the vendor's location. The courier acts as the neutral referee—verifying that the item physically exists and matches the classification before checking it in. The courier's internal dispatch system fires a webhook updating Luggik's state machine to `IN_TRANSIT`.

5 **Automated Split-Settlement**

Upon successful doorstep drop-off, the courier flags the item as delivered. Luggik instantly consumes this delivery webhook and triggers parallel transfers through Nomba: 95% of the item cost routes straight to the vendor, and the delivery fee routes to the logistics partner. The burner account is instantly liquidated and deleted.

Why Luggik Wins: Competitive & Hackathon Advantage

Luggik redefines digital escrow by making payments context-aware and deeply unified with real-world infrastructure. By removing subjective disputes, we build a highly predictable payment clearing-house.

STRATEGIC METRIC	TRADITIONAL ESCROW APPS	LUGGIK LAYER (NOMBA POWERED)
Adoption Friction	High. Requires both buyer and seller to download apps and complete KYC.	**Zero.** Invisible to the vendor; buyer uses a clean 1-minute web checkout interface.
Dispute Processing	Manual. Support agents spend hours tracking screenshots and video links.	**Automated.** Resolved instantly by state-verified logistics status webhooks.
Merchant Security	Low confidence in new, unproven escrow web portals.	**High.** Branded and secured natively by Nomba's banking rails.
Capital Velocity	Stuck. Capital stays locked inside closed wallets for extended days.	**Instant.** Automated settlement happens via real-time webhooks.

Nomba API Integration Architecture

Luggik does not treat Nomba as a passive payment gateway; it utilizes Nomba's core structural APIs as the underlying infrastructure primitives to drive automated, real-time value routing.

- **Virtual Accounts API:** Dynamically generates ephemeral, unique account endpoints for every single order, serving as isolated programmatic holding cells for transaction funds.
- **Webhooks & Transaction API:** Listens to instant settlement notifications, triggering automated B2B logistical queues the split-second cash drops from the consumer's traditional banking channel.

- **Transfers API:** Executes complex, programmatic atomic split-payouts upon completion of the physical delivery, ensuring zero-latency compensation for both merchants and transport agencies.

Judges' Q&A Defense & Scrutiny Mitigation

Q: What happens if a dispatch rider goes rogue or colludes with a party?

A: Luggik anchors its trust directly into the enterprise logistics partner platforms (e.g., GIGL, Topship) via secure, corporate API keys, not individual riders' personal phones. The logistics networks carry full corporate insurance and liability for tracking statuses marked inside their systems. If an internal system hack or incorrect check-in occurs, the legal and financial accountability remains bound to the carrier's corporate SLA, completely shielding Luggik and Nomba from liability.

Q: Why would a logistics aggregator or vendor agree to this ecosystem?

A: Logistics companies desperately want this. It completely eradicates the high costs of "Return to Sender" (RTS) issues caused by consumers backing out of Pay-on-Delivery collections at the final hour. For vendors, it guarantees that 100% of their orders are backed by verified liquid cash before their inventory leaves the shelf.